

Elijah Olson

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Education

B.S. Computer Science & Engineering | Minor in Philosophy - *University of California, Merced* ————— 05/2026

- **Relevant Coursework:** Deep Learning, Neural Networks, Natural Language Processing, Algorithm Design and Analysis, Data Structures and Design Patterns, Full-stack Web Development, Natural Image Processing, Tech Ethics, Philosophy of the Human Mind, Cognitive Neuroscience, Contemporary Ethics, Metaphysics, Modern Philosophy, Ancient Philosophy, Calculus I-III, Linear Algebra, Linear Analysis, Discrete Mathematics, Statistics

Experience

Website Manager and Data Technician | *University of California, Merced* ————— 09/2025 - Current

- Completed entire **full-stack** redesign of the [CEC website](#), improving accessibility and user experience to achieve 22% increase in engagement and 45% increase in event attendance.
- Designed and implemented surveys, experience reports, and automated **data pipelines** for all UC Merced-affiliated community programs, creating streamlined feedback and engagement tracking.

Neural Simulation Architect | [SIMBRAIN](#) ————— 06/2025 - 08/2025

- Transformed theoretical neural circuit models into fully functional implementations in **Kotlin & Java**, enabling researchers to simulate diverse neurological systems across Simbrain's platform.
- Enhanced platform accessibility through library improvements, comprehensive documentation, and **UX** refinements, enabling neuroscientists and students of all skill levels to build functional neural models.

Fullstack Engineer Intern | *Patient Safety* ————— 06/2024 - 08/2024

- Architected backend **data pipelines** in **Python** and **SQL**, automating CSV ingestion with validation, creating infrastructure for medical data management across millions of records.
- Collaborated with full-stack engineers on **Django** and **PostgreSQL** interfaces design to optimize seamless backend-frontend integration, and prioritize **UX** through the use of **Google Sprites**.

Research Assistant | *University of California, Merced* ————— 11/2023 - 03/2024

- Developed data processing scripts in **Python** to extract, clean, and organize large sets of EEG data, identifying and isolating relevant neural features to create structured datasets and visualizations with **Pandas** and **Matplotlib**.
- Collaborated with a graduate researcher to translate processed neurological data into actionable insights about sensory processing in the human brain.

Software Engineer Intern | [Tappity Inc.](#) ————— 04/2020 - 10/2021

- Collaborated with full-stack engineers using **Python**, **JavaScript**, and **React** to design and implement new experiential learning activities for the science education app, leveraging knowledge from prior child care experience, and ultimately boosting user engagement and information retention among K-8 students.

Projects

- [Syllabud.app](#): live website and second-place hackathon winner. **Full-stack AI** course creator using **Google Gemini**, **FastAPI**, **React**, **PostgreSQL**, and **CSS** with **JWT** authentication. Users input subject, timeline, depth, and learning style to generate personalized curricula with curated resources. Deployed on **Railway**.
- [Function Visualization](#): A design tool made in **Python** and **Pygame** that renders complex fractals (Mandelbrot, Fibonacci, etc.) by mapping functions to visual outputs, exploring the intersection of math and generative art.

Skills

- **Programming & Dev:** C++, C#, Python, Kotlin, JavaScript, HTML, GDScript, Django, React, SQL, Flask, Pandas, TensorFlow, PyTorch, GitHub, PostgreSQL
- **AI & Neuro:** Deep Learning, ML, Neural Networks, Comp Neuroscience, Neural Sim, Hidden-Layer Analysis
- **Philosophic:** Tech Ethics, Philosophy of Mind and Cognition, Ethical AI systems, Contemporary and Modern Ethics
- **Procedural:** Use of AI, Project Engineering, UX, Data Visualization, Technical Documentation, Cross-Functional Communications, Systems Thinking, Human-Centered Design, Figma, Lucid